



Föreläsning: 8

Digitalkamera och bildlagring

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Digitala kameran



Digitalkameran i var mans hand!

Igen: Viktigaste tekniken?

Olika sensorer?

FLASH-minnen

Olika bildformat?

TIFF, GIF, JPEG,...



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Digital Cameras



no "film" – a sensor converts light into electrical signals which can be stored



The sensor:

- CCD – Charge Coupled Device
- CMOS - Complementary Metal Oxide Semiconductor

Both are solid state sensors.

- CCD - higher light sensitivity
- CMOS - lower power and lower price

The data is then (usually) stored in a solid-state flash memory, e.g. SmartMedia cards, CompactFlash and Memory Stick

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The Sensor



The sensor converts light into electrical signals, compare to a solar cell:



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CCD or CMOS

- CCD – captured charge is transported across the sensor
 - CMOS - transistors located next to pixel. Amplification and more traditional signaling.
 - The transistors close to the pixel in CMOS reduce light sensitivity, i.e. photons hit the transistors instead of the pixel.
 - CMOS traditionally consumes less power. CCDs can consume as much as 100 times more power than an equivalent CMOS sensor.
 - CMOS chips can be fabricated in standard silicon CMOS, so they tend to be inexpensive compared to CCD sensors.
- “Old statements” ?
- CCD sensors have been mass produced for a longer period of time, so they are more mature. They tend to have higher quality and more pixels
 - CCD sensors create high-quality, low-noise images. CMOS sensors, traditionally, have more noise.

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Profesional digital SLR cameras

Canon EOS-1Ds MkII



US\$ 7 000 (body only)
4992 x 3328 pixel
16.6 million eff. pixels
CMOS sensor
Compact Flash (Type I or II), SD card
4 fps max 32 JPEG, 11 RAW

Nikon 2DXs



US\$ 4 500 (body only)
4288 x 2848 pixel
12.2 million eff. pixels
CMOS sensor
Compact Flash (Type I or II)
5 fps

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Ultra-Compact Digital Cameras

Leica D-LUX 3



US\$???
4224 x 2376 pixel
10.2 million eff. pixels
CCD sensor
106 x 56 x 26 mm

Canon IXUS 900 Ti



US\$ 500
3648 x 2736 pixel
10.0 million eff. pixels
CCD sensor
91.2 x 59.6 x 28.2 mm

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Some typical differences

	Digital SLR camera	Digital Compact Camera
Electronic viewfinder	NO	YES
Capture movies	NO	YES
Shutter lag	Small/unnoticeable	Can be considerable
Sensor size (area)	Large - low noise	Small - high noise
Image buffers	Large & fast	Small & Slow
Lenses	Can be exchanged	Fixed
RAW format	YES	NO

MEM. MONEY

ENERGY

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Digital Cameras in mobile phones



3.2M Pixels

Is there higher out there?



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Autofocus

- Active
 - measure distance
 - ultra sound (first)
 - infra red (works in the dark)
- Passive
 - image processing
 - the lens is moved back and forth by a motor to find focus
 - Only horizontal, or in both directions
 - Doesn't work on "homogeneous" surface

Out of focus



In focus



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Some other electronic cameras ...

Infrared surveillance camera



Network (web) camera



Thermal imaging camera



Wireless color video spy-cam



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... and the ultimate ...

Wireless teddy bear spy cam



In the ad: "This unique product make an ideal video baby monitor"

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Digital storage of images

Size: 370 x 477 pixels



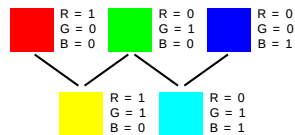
Loss-less storage:
The color each pixel is represented by the amount of three colors

RGB - red/green/blue
(monitors)

or

CMY - cyan/magenta/yellow
(printing)

Examples:



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Digital storage of images

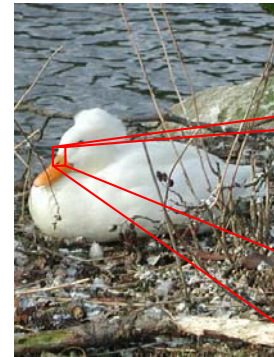
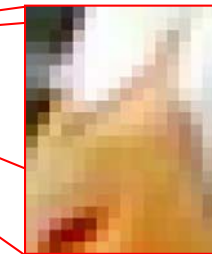


Image compression:

Each pixel is represented by three values, as in loss-less storage, but we can decrease the "file" size by



reducing the total number of colors

and/or

explore the fact that neighbouring pixels are of almost the same color.

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Digital storage of images

Still images (some formats):

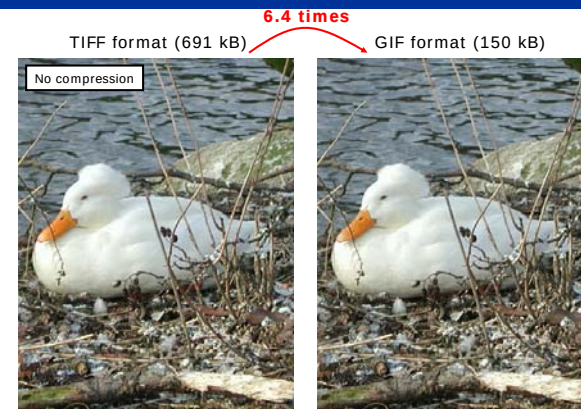
- TIFF** - Tag Image File Format
A "high quality" storage format. Does not compress the images (it is optional) and is frequently used by the graphics industry.
- GIF** - Graphics Interchange Format
An early compression format (1980's) which does loss-less compression. Is however limited to a maximum of 256 different colors in an image.
- JPEG** - Joint Photographic Experts Group Interchange Format
A common compression format (1990's) for primarily photographic images. Does lossy compression (variable). Very common on web-pages and in digital cameras.

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TIFF vs. GIF

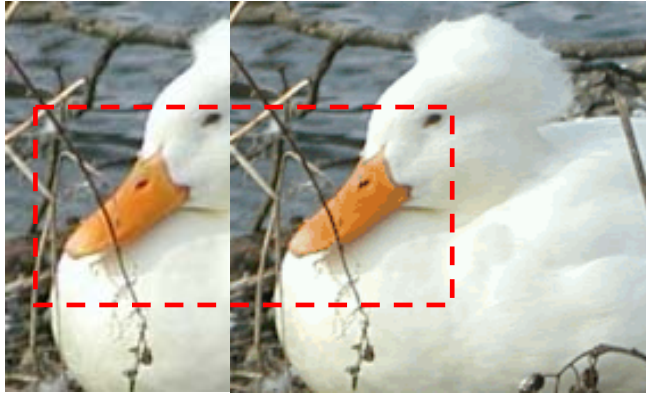


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TIFF vs. GIF (close up)

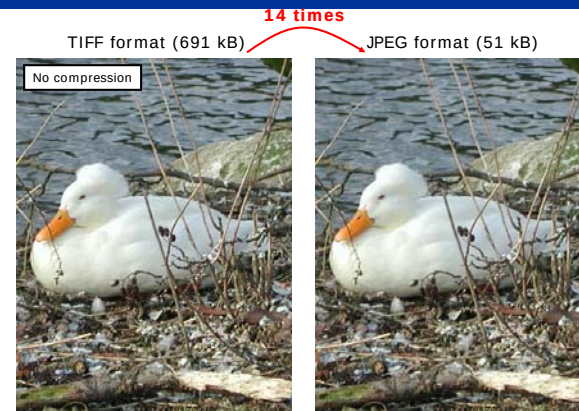


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TIFF vs. JPEG

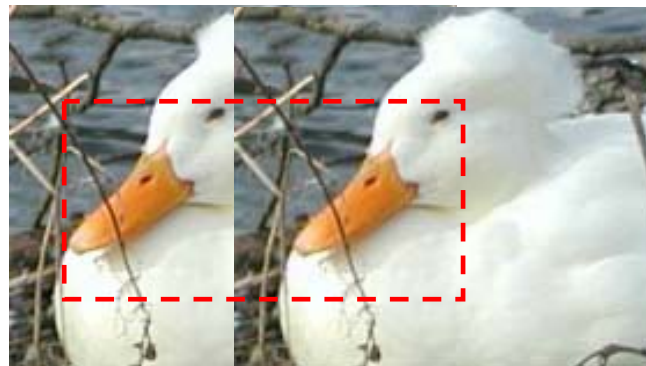


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TIFF vs. JPEG (close up)

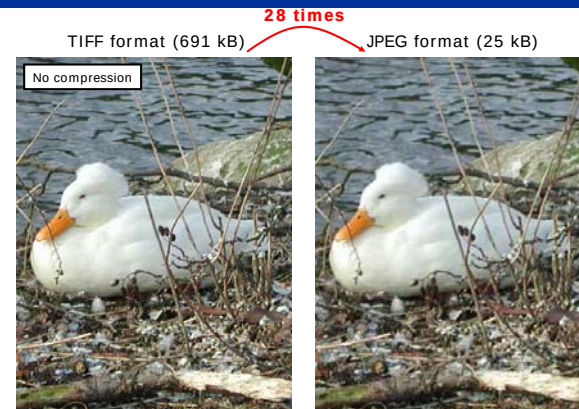


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TIFF vs. JPEG

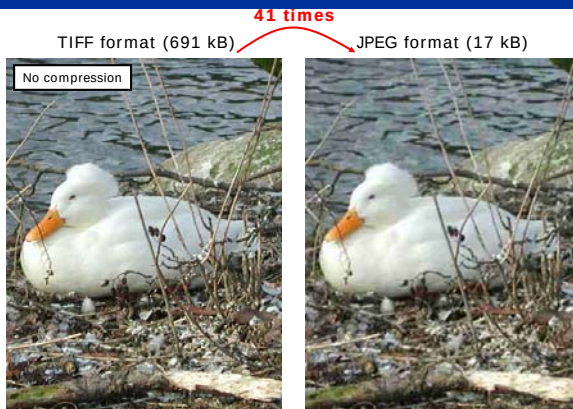


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TIFF vs. JPEG

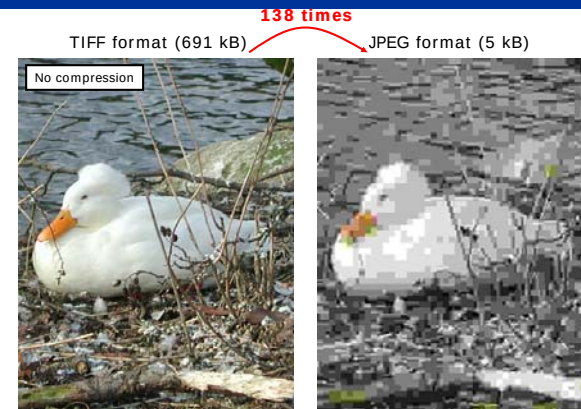


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TIFF vs. JPEG



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Storage

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Digital Cameras Storage

Solid-state flash memory, e.g:

- SmartMedia cards
- CompactFlash
- Memory Stick



CD



Hard drive



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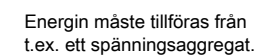
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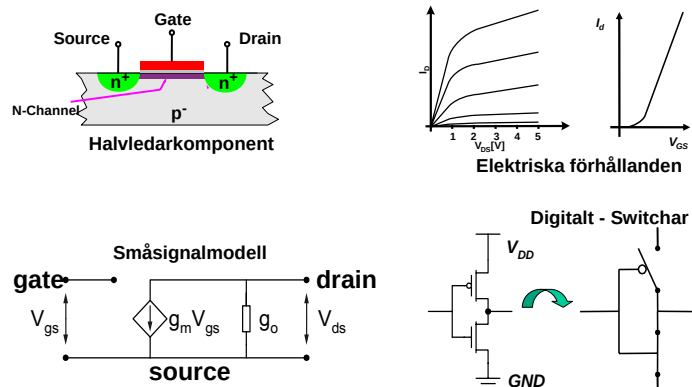


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Vad är en transistor?

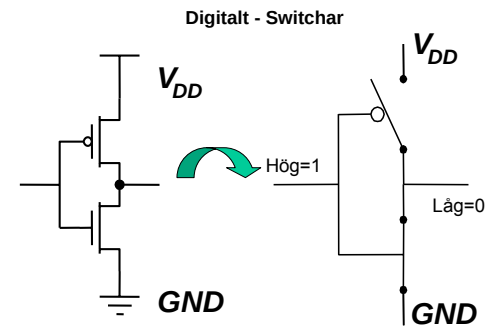


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The Transistor as a Switch

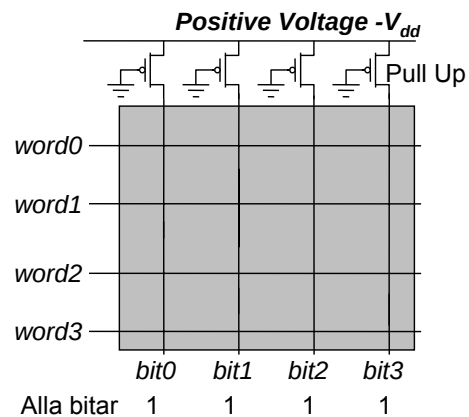


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ROM – Read Only Memory

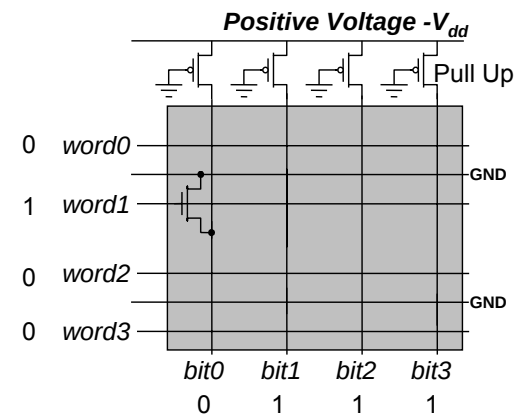


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ROM – Read Only Memory

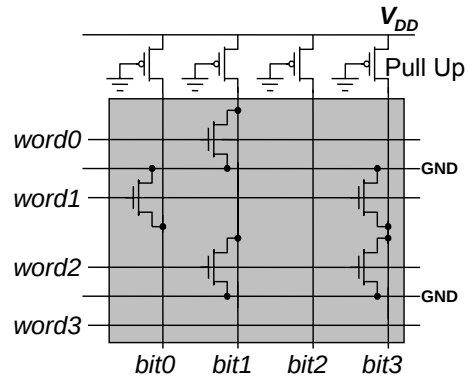


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ROM – Read Only Memory



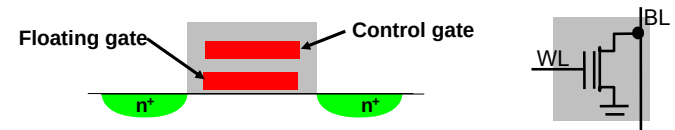
Where transistors are placed decide the memory content.

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Flash memories – Floating gate transistors



Floating gate is left unconnected

- If charged heavily negative $\Rightarrow$ No channel $\Rightarrow$ No Transistor
- If charged removed $\Rightarrow$ Channel $\Rightarrow$ Transistor

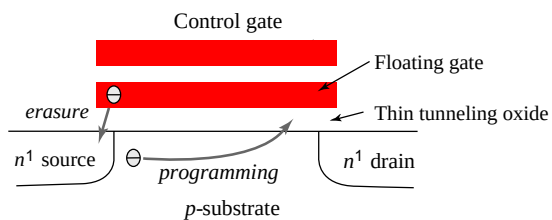
• EPROM, EEPROM and Flash has different ways of controlling the charge of the floating gate

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Flash EEPROM



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NOR and NAND Flash

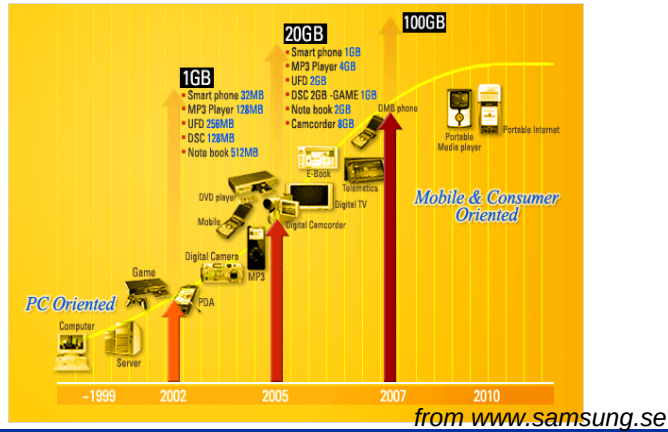
- NOR
 - fast read access
 - slow erasure and programming are slow
 - Samsung: 512Mbit/166MHz (from www.samsung.com)
- NAND
 - slower read access
 - larger storage density
 - fast erasure and programming
 - Samsung: 32 Gbit/20MHz (from www.samsung.com)

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Living in the stage of 20GB memory after passing through the dark-age of 1GB in 2002...

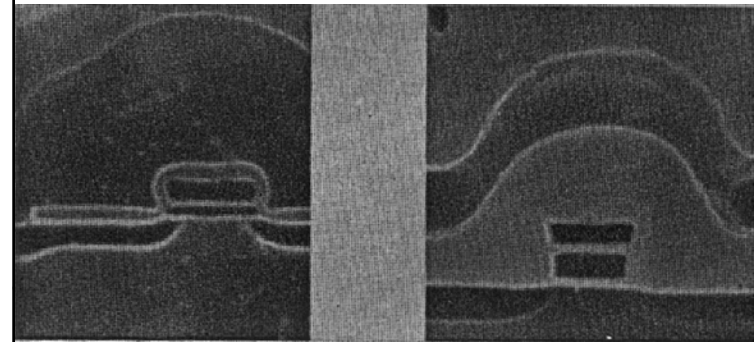


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Cross-section



Flash

Courtesy Intel

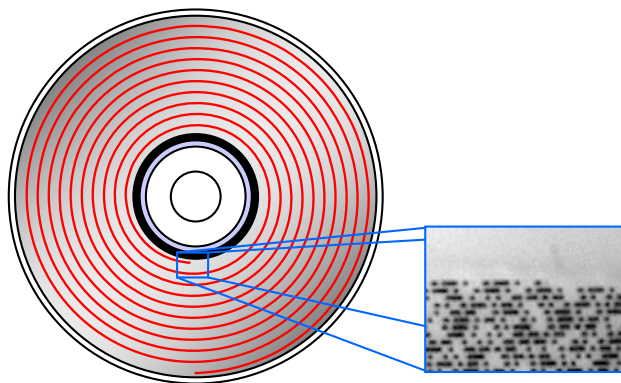
EPROM

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Compact disk surface



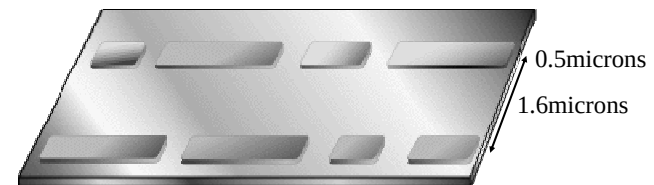
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CD - Compact Disc

If you could stretch the track off a CD into a straight line, it would be almost 5 km.



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LaserDisc

Same type of spiral as the CD and with the same pitch distance of $1.6 \mu\text{m}$, but the video signal is analog.

The length of bumps and lands represent **analog** signal amplitude.

The spiral on a 12 inch LaserDisc is about 40 km long.



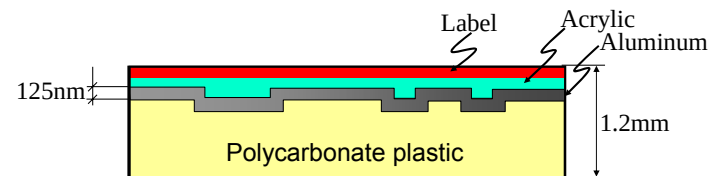
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The CD

- Sample rate : 44.1 ksamples/second
- Wordlength: 16 bits (2 bytes)
- Information on CD:
 - 44,100 samples/channel/second x
 - 2 bytes/sample x 2 channels x
 - 74 minutes x 60 seconds/minute =
 - 783,216,000 bytes

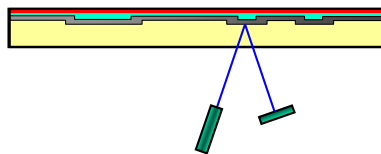


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Reading the CD



Bumps and pits reflect light differently, thus 1s and 0s.

Tracking makes sure that the laser hits the center of the "digit" by adjusting the speed, e.g. three detectors.

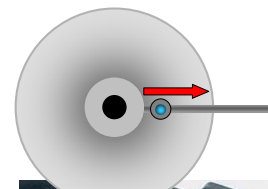
Data is encoded to avoid long stretches without bumps which would make the laser lose track.

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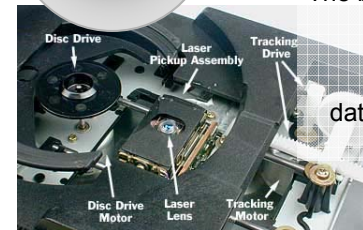
The CD player



As the laser moves outwards, the speed is reduced from 500rpm to 200 rpm

The bumps travel past the laser at a constant speed

data comes off the disc at a constant rate



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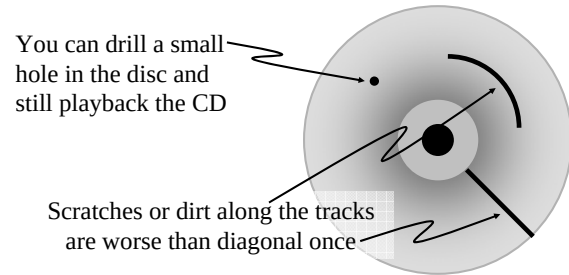
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Error correcting codes

The data is coded so that if there is a "limited" error, they error can be corrected

Error correcting coding will increase the number of bits.

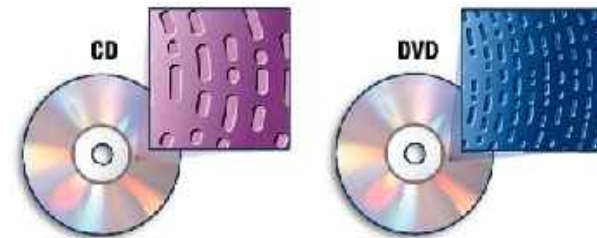


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CD vs. DVD (Digital Versatile Disc) first Digital Video Disc



Track Pitch: 1.6μm
Bump width: 0.5μm

0.74μm
0.32μm

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CD vs. DVD audio format

Specification	CD Audio	DVD Audio
Sampling Rate	44.1 kHz	192 kHz
Samples Per Second	44,100	192,000
Sampling Accuracy	16-bit	24-bit
Number of Possible Output Levels	65,536	16,777,216

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DVD

Single-sided, single layer (4.38GB)



Movie
2 hours

Single-sided, double layer (7.95GB)



Movie
4 hours

semi-reflective gold layer, laser can focus through the outer layers.

Double-sided, double layer (15.9GB)



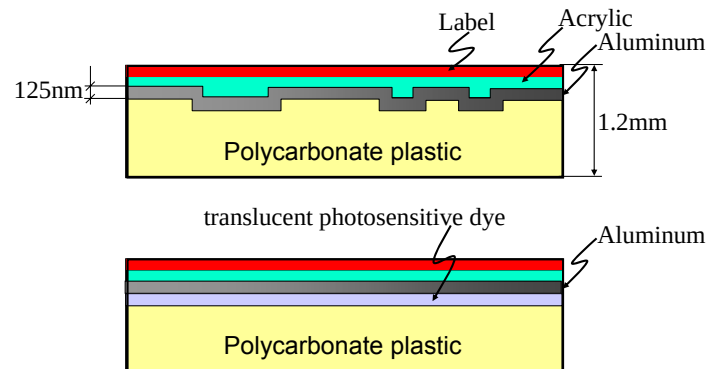
Movie
8 hours

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CD-Recordables - CD-Rs

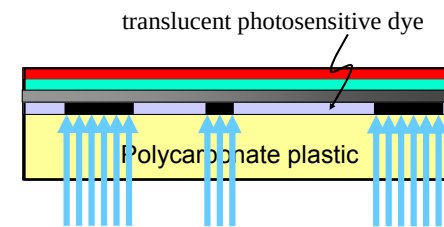


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CD-Recordables - CD-Rs



When the translucent photo sensitive dye gets heated by light it darkens and wont let the light through.
By doing so a pattern of 1s and 0s is created.

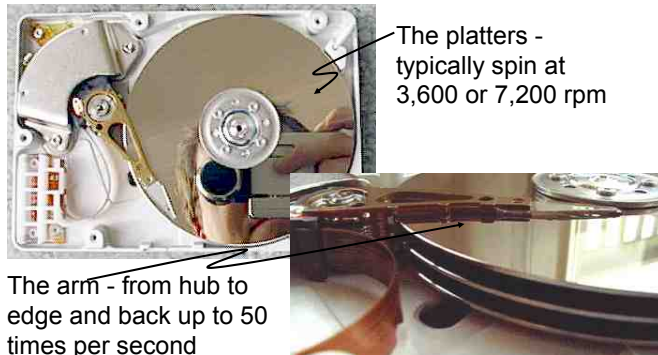
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Hard drive

magnetic recording - easily erased and rewritten



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Hard drive



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Nästa föreläsning



- Batterier och strömförsörjning

- Energi och effekt
 - förbrukning
 - generering
- Batterier
- Bränsleceller
- Solceller
- Seebeck
- Peltier
- Mekanisk